Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 3	Pro		Civic and Social Service Internship	0	0	0	0	1
	BSC		Mathematics – 3	3	1	0	4	4
	ESC		Introduction to Artificial Intelligence	3	0	0	3	3
	PC		Building Materials and Construction Technology	3	0	0	3	3
	PC		Building Planning and Computer Aided drawing	1	0	2	3	2
	PC		Solid Mechanics	4	0	0	4	4
	PC		Concrete Technology	2	0	0	2	2
	PC		Material Testing Laboratory	0	0	2	2	1
				16	**1**	**4**	**23**	**20**
Semester 4	ESC		Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC		Industry 4.0 - Lab (Department Specific)	0	0	2	2	1
	OE		Open Elective 1 (From Other School)	3	0	0	3	3
	PC		Fluid Mechanics	3	0	2	5	4
	PC		Engineering Geology and Soil Mechanics	3	0	2	5	4
	PC		Surveying Practices	3	0	2	5	4
	PC		Structural Analysis	3	1	0	4	4
	Pro		**Industrial Orientation**					0
				17	**1**	**8**	**26**	**22**

					Industry 4.0					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand Industry 4.0 concepts and technologies for transforming industrial processes.
2. Apply BIM principles and tools in civil engineering projects, integrating with GIS, IoT, and project management.
3. Evaluate and apply sustainable building design principles in the context of Industry 4.0.
4. Analyze emerging technologies in civil engineering for enhanced project efficiency and sustainability.

UNIT I : INTRODUCTION AND BASIC CONCEPTS OF INDUSTRY 4.0 **7 Hrs.**

Introduction to Industry 4.0, Definition, General framework, Application areas, Introduction to the Evolution of Industrial revolutions, Key features, Need and benefits, Introduction to Industry 4.0 core technologies: Big data, Advanced Robotics, Simulation, Integration, Internet of Things (IoT), Artificial Intelligence (AI), Cybersecurity, Cloud computing, Additive manufacturing and Augmented Reality, Overview of machine learning for Industry 4.0, Transformation of industrial processes through the integration of modern technologies such as sensing and actuation, communication, and computational processing..

UNIT II : BUILDING INFORMATION MODELLING APPLICATIONS **7 Hrs.**

Introduction to Building Information Modelling (BIM), Principles and Concepts of BIM in Project Management, BIM Software and Tools for Project Planning, Execution, and Monitoring, BIM Applications in Civil Engineering Projects, BIM Level of Detail (LOD) and Level of Development (LOD) Standards, Best Practices for BIM Implementation in Projects, Integration of BIM with GIS and IoT, Project Management Principles in Civil Engineering, BIM for Project Planning and Scheduling, BIM for Project Tracking and Monitoring.

UNIT III : SUSTAINABILITY AND CONSTRUCTION INNOVATION **7 Hrs.**

Introduction to Sustainable Building Design and its Alignment with Industry 4.0 Principles, Principles and concepts of sustainability in construction, Sustainable building materials and techniques, Energy-efficient design strategies, Water conservation and management in building design, Waste management and recycling in construction, Green building certifications and standards (e.g., LEED, BREEAM, GRIHA), Indian building codes and regulations for sustainable construction, Simulation tools for sustainable building design (e.g., EnergyPlus, Designbuilder, IES VE), 3D Printing in Construction for Sustainable Practices.

UNIT IV: EMERGING TECHNOLOGIES IN CIVIL ENGINEERING **7 Hrs.**

Internet of Things (IoT) Applications, Augmented Reality (AR) and Virtual Reality (VR) in Civil Engineering, Robotics and Automation, Use of Drones (UAVs), Smart Sensors and Monitoring Systems, Advanced Materials, Remote Sensing Applications, Drone Technology for Surveys and Monitoring, Smart Cities, Case Studies and Future Trends

TOTAL HOURS: 28 Hrs.

COURSE OUTCOMES:

CO-1: Demonstrate an understanding of Industry 4.0 technologies and their impact on industrial processes.

CO-2: Apply Building Information Modelling (BIM) principles and tools in civil engineering projects.

CO-3: Evaluate sustainable building design principles and apply them in construction projects.

CO-4: Integrate technology and data management strategies for efficient civil engineering project execution.

CO-5: Implement innovative technologies to enhance civil engineering project efficiency.

CO-6: Utilize advanced tools for effective monitoring and management of civil engineering projects.

TEXT-BOOK AND REFERENCE BOOKS:

1. Sanjay Sharma, **"Introduction to Industry 4.0: A Practical Approach to Understanding Industry 4.0 Concepts and Technologies"**, CRC Press, 2019
2. Karen Kensek, **"Building Information Modeling (BIM) in Civil Engineering"**, Wiley, 2017
3. Charles J. Kibert, **"Sustainable Construction: Green Building Design and Delivery"**, Wiley, 2016
4. Rajat Gupta, **"Emerging Technologies in Civil Engineering: Advances in Remote Sensing, IoT, and Robotics"**, Springer, 2020

Teaching Scheme					Industry 4.0 Laboratory					
					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand aapplications of Industry 4.0 technologies through case studies and videos.
2. To learn BIM software for project design, practice 2D and 3D rendering techniques.
3. To learn QGIS software for remote sensing applications like land cover mapping.
4. To gain hands-on experience in drone flight and control and data analysis.

UNIT I : INTRODUCTION AND BASIC CONCEPTS OF INDUSTRY 4.0 **4 Hrs.**

Case Studies and Videos on Big data, Advanced Robotics, Simulation, Integration, Internet of Things (IoT), Artificial Intelligence (AI), Cybersecurity, Cloud computing, Additive manufacturing, Augmented Reality.

UNIT II : BUILDING INFORMATION MODELLING APPLICATIONS **6 Hrs.**

BIM software and tools for project design, 2 D, 3D Rendering. Application of BIM in civil engineering projects, including project planning, scheduling, and tracking. Case-based learning about building materials and techniques. Simulation exercises using tools like EnergyPlus and Designbuilder for sustainable building design, Incorporating innovative construction materials for sustainability.

UNIT III: EMERGING TECHNOLOGIES IN CIVIL ENGINEERING **4 Hrs.**

QGIS software for remote sensing applications like land cover mapping and climate change analysis, Demonstrations of drone flight and control for data collection and analysis in civil engineering projects.

TOTAL HOURS: 14 Hrs.

COURSE OUTCOMES:

CO-1: Understand and apply Industry 4.0 technologies through case studies and videos.

CO-2: Utilize BIM software for project design, including 2D and 3D rendering.

CO-3: Implement BIM in civil engineering projects for planning and tracking.

CO-4: Use simulation tools for sustainable building design and innovative materials.

CO-5: Apply QGIS for remote sensing applications like Land Cover Mapping and Climate Change Analysis.

CO-6: To learn about drones for data collection in civil engineering.

TEXT-BOOK AND REFERENCE BOOKS:

1. Alasdair Gilchrist, **"Industry 4.0: The Industrial Internet of Things"**, Wiley, 2016.
2. Willem Kymmell, **"Building Information Modelling: Planning and Managing Construction Projects with 4D CAD and Simulations"**, McGraw-Hill, 2008.
3. Paul Bolstad, **"GIS Fundamentals: A First Text on Geographic Information Systems"**, Eider Press, 2016.
4. Micahel Chasen, **"Drone Mapping for Civil Engineering"**, CRC Press, 2020.

					Fluid Mechanics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals concepts of fluid mechanics
2. To established relevance of fluid mechanics with civil engineering
3. To learn the concepts of fluid and its properties
4. To learn the concepts of fluid flows and properties

UNIT I: FUNDAMENTALS OF FLUID MECHANICS **14 Hrs.**

Properties Of Fluid: Introduction of Fluid Mechanics, Properties of Fluid – Density, Specific Gravity, Viscosity, Compressibility, Bulk Modulus, Surface Tension, Capillarity. Pressure Measurement: Pascal's Law, Pressure Measurement – Manometers. Hydrostatic Forces: Total Pressure and Centre of Pressure – Vertical Plane surface, Horizontal Plane Surface, Inclined Plane Surface, Curved Surface and Lock Gates. Buoyancy: Buoyancy, Metacentre and Metacentric Height - Determination, Equilibrium Conditions

UNIT II: FLUID KINEMATICS AND DYNAMICS **14 Hrs.**

Fluid Kinematics: Introduction, Types of Flow, Continuity Equation – Three Dimensions and Cylindrical Co-ordinates, Velocity and Acceleration, Velocity Potential Function and Stream Function, Vortex Flow. Fluid Dynamics: Introduction, Euler's and Bernoulli's Equation, Practical Application of Bernoulli's Equation – Venturimeter, Orifice Meter and Pitot Tube, Momentum Equation, Free Liquid Jets.

UNIT III: VISCOUS AND TURBULENT FLOW **14 Hrs.**

Viscous Flow: Introduction, Flow of Viscous Fluid – Circular pipe and Parallel Plates, Correction Factors, Power Absorbed in Viscous Flow, Loss of Head due to Viscous Flow, Coefficient of Viscosity. Turbulent Flow: Introduction, Reynolds Experiment, Friction Loss in Pipe Flow, Shear Stress in Turbulent Flow.

UNIT IV: PIPE FLOW **14 Hrs.**

Introduction, Energy Loss due to Friction, Minor Energy Losses, HGL and TEL, Flow Through Syphon, Flow Through Pipes – Series, Compound, Equivalent, Parallel, Pipe Network – Hardy Cross Method.

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES:

CO1 : Remember the Fundamental Concepts of Fluid Mechanics.
CO2 : Understanding of the Behavioural Aspects of Fluid in Different Conditions.
CO3 : Understanding and Classification of Fluid in Various Categories.
CO4 : Develop a Relationship Between Properties of the Fluid.
CO5 : Apply and Solve the Numerical Solution for the Problems of Fluid Mechanics.
CO6 : Compare and Conclude the Aspect of Fluid Mechanics for Problem Solving Skills.

TEXT-BOOK AND REFERENCE BOOKS:

1. Dr. R K Bansal, **"A textbook of fluid mechanics and hydraulic machines"**, Laxmi Publishers, New Delhi, 2015.
2. Dr. A K Jain, **"Fluid Mechanics including Hydraulic Machines"**, Khanna Publishers, New Delhi, 2015.
3. R K Rajput, **"A textbook of fluid mechanics"**, S. Chand Publishers, 2013.
4. S Ramamrutham, **"Hydraulics fluid mechanics and fluid machines"**, Dhanpat Rai Publishers, New Delhi, 2015.
5. P N Modi and S M Seth, **"Hydraulics and fluid mechanics"**, Standard Book House Publishers, New Delhi, 2015.
6. R J Garde and A G Mirajgaoker, **"Engineering fluid mechanics"**, Scitech Publishers, New Delhi, 2012.

Teaching Scheme					Fluid Mechanics Laboratory					
					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To learn the fundamentals concepts of fluid mechanics.
2. To established relevance of fluid mechanics with civil engineering.
3. To learn the concepts of fluid and its properties.
4. To learn the concepts of fluid flows and properties.

LIST OF EXPERIMENTS

1 Viscosity measurement using Efflux Time for Oil (Only One Liquid)
2 Application of Reynolds Experiment for Identification of Flow
3 Application of Venturimeter and Orificemeter for Flow Measurement
4 Determination of Metacentric Height for Floating Body using Experimental Setup
5 Calculation of Head Loss (Friction Loss) for the Close Conduits
6 Determination of the Minor Losses incurred in the Pipe Assembly
7 Experimental Verification of Bernoulli's Theorem
8 Measurement of Flow Discharge in Open Channel Flume using Pitot Tube
9 Evaluate the Hydraulic Co-efficient for Orifice and Mouthpiece
10 Field Visit and Application of Experiment for Pipe Network Distribution

COURSE OUTCOMES:

CO1 : Relate the Concepts of Fluid Mechanics for Understanding the Concepts
CO2 : Apply the Concept of Fluid Mechanics for Verification of the Experimental Data
CO3 : Illustrate the Concepts for Friction Losses in Fluid Mechanics
CO4 : Examine the Experimental Data for Analysis of Losses within Experimental Set-up
CO5 : Analyse the Fluid Properties Using Different Aspect and Methods
CO6 : Conclude the Fluid Mechanics Concepts for Evaluation of the Field Data

TEXT-BOOK AND REFERENCE BOOKS:

1. Dr. R K Bansal, **"A textbook of fluid mechanics and hydraulic machines"**, Laxmi Publishers, New Delhi, 2015.
2. Dr. A K Jain, **"Fluid Mechanics including Hydraulic Machines"**, Khanna Publishers, New Delhi, 2015.
3. R K Rajput, **"A textbook of fluid mechanics"**, S. Chand Publishers, 2013.
4. S Ramamrutham, **"Hydraulics fluid mechanics and fluid machines"**, Dhanpat Rai Publishers, New Delhi, 2015.
5. P N Modi and S M Seth, **"Hydraulics and fluid mechanics"**, Standard Book House Publishers, New Delhi, 2015.
6. R J Garde and A G Mirajgaoker, **"Engineering fluid mechanics"**, Scitech Publishers, New Delhi, 2012

					Engineering Geology and Soil Mechanics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn different elements of Engineering Geology such as Mineralogy, Petrology and Structural Geology.
2. To impart knowledge to classify the soil based on index properties subsequently find their engineering properties.
3. To understand the fundamental concepts of compaction, permeability, stress transformation, stress distribution, consolidation and shear strength of soils.

UNIT I: FUNDAMENTALS OF ENGINEERING GEOLOGY **14 Hrs.**

Definition and scope of geology –Geological Timeline-Geological Earth Processes - Study of the rock forming minerals, properties - Classification of rocks - Description - Occurrence - Engineering properties - Structural geology: Dip, strike, folds, faults and joints-their engineering aspects - Case Studies

UNIT II: SOIL BASICS, INDEX PROPERTIES AND SOIL CLASSIFICATIONS **14 Hrs.**

Soil formation - Clay mineralogy - Surface activity - Volume-weight relationships - Grain size distribution - Sieve analysis and hydrometer analysis - Consistency limits and their determination - Soil classification systems

UNIT III: COMPACTION, CONSOLIDATION AND STRESS DISTRIBUTION **14 Hrs.**

Compaction – Moisture - density relations - Factors affecting compaction - Field compaction methods - Field compaction control - primary consolidation concept - void ratio - pressure curve - Field curve - Laboratory test - Vertical Stress distribution in soil - Boussinesq equation - line load - Uniformly distributed loads - Influence chart - approximate methods- Westergaard's equation - Pressure bulb

UNIT IV: PERMEABILITY, SEEPAGE AND SHEAR STRENGTH **14 Hrs.**

One dimensional flow through soil - Permeability - Darcy's Law - Field and laboratory permeability tests - Flow through stratified soil - Seepage - Two dimensional flow - Laplace's equation – Introduction to flow nets - Shear Strength of soil - Importance - Mohr Coulomb's Strength theory - Laboratory and field tests - Factors affecting shear strength - Types of shear tests based on drainage condition

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES:

CO-1 : Explain geological earth processes in formation of soils, rock, mineralogy, structural features
CO-2 : Develop interrelations between index properties of soils
CO-3 : Classify the soil based on Index properties of soils
CO-4 : Apply principles of compaction and consolidation to the control field conditions
CO-5 : Demonstrate the concept of stress distribution and shear strength of soils
CO-6 : Compute permeability and seepage for different soils and hydraulic structures

TEXT-BOOK AND REFERENCE BOOKS:

1. BC Punmia, Ashok Kumar Jain, Arun Jain, **"Soil Mechanics and Foundations"**, Laxmi Publication.
2. KR Arora, **"Soil Mechanics and Foundation Engineering"**, Standard Publisher Dist.
3. Parbin Singh, **"Engineering and General Geology"**, S.K. Kataria & Sons.
4. Shashi Gulathi, Manoj Dutta, **"Geotechnical Engineering"**, Tata Mcgraw Hill Publisher.

					Engineering Geology and Soil Mechanics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To discriminate rock forming minerals and different types of rocks
2. To interpret topographical and geological map
3. To determine index and engineering properties of soils

LIST OF EXPERIMENTS

1. Study and identification of rocks and rock forming minerals
2. Determination of moisture content and specific gravity of fine-grained soils
3. Determination of field density using core cutter method
4. Determination of field density using sand replacement method
5. Determination of liquid limit and plastic limit of soils
6. Grain size analysis of soils
7. Determination of maximum dry density and optimum moisture content using standard Proctor test
8. Determination of relative density of sandy soils
9. Determination of permeability of soils using (i) constant head (ii) falling head permeability test
10. Determination of compressibility of soil using consolidation test
11. Determination of shear strength parameters of soil by direct shear test
12. Determination of unconfined compressive strength of soil
13. Lab based project

COURSE OUTCOMES:

CO-1: Identify minerals and rock
CO-2: Determine field density and compaction properties of soil
CO-3: Classify the soil based on the index properties and physical characteristics
CO-4: Determine the shear strength parameters of soil
CO-5: Determine the permeability of soil
CO-6: Evaluate the compressibility characteristics of soil

TEXT-BOOK AND REFERENCE BOOKS:

1. Laboratory Manual
2. BC Punmia, Ashok Kumar Jain, Arun Jain, **"Soil Mechanics and Foundations"**, Laxmi Publication, 16th Edition, 2015.
3. KR Arora, **"Soil Mechanics and Foundation Engineering"**, Standard Publisher Dist., 2009.
4. Parbin Singh, **"Engineering and General Geology"**, S.K. Kataria & Sons, 2014.
5. Shashi Gulathi, Manoj Dutta, **"Geotechnical Engineering"**, Tata Mcgraw Hill Publisher, 2009.

					Surveying Practices					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	5	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the principles and procedure for linear, angular measurement and levelling
2. To study the various advanced surveying techniques for civil engineering.
3. To understand area and volume measurement with surveying techniques.
4. To utilization of Remote sensing, GIS techniques in construction activities.

UNIT-1: INTRODUCTION BASIC SURVEYING **14 Hrs.**

Linear survey, linear measurements using chains and tapes, chaining and ranging errors and corrections, procedure of field work, angular survey introduction, WCB and QB, computation of angles from bearings, local attraction and corrections, principle of levelling, methods of levelling, curvature and refraction corrections, contouring, methods of contouring, uses of contour maps.

UNIT-2: TRIANGULATION AND CURVE SETTING **14 Hrs.**

Theodolite survey: Instruments, Measurement of horizontal and vertical angle; Horizontal and vertical control – methods. Elements of simple, compound, and Reverse curves, Method of setting out simple circular curve, Computation of area of regular and irregular boundary computation of volume for earthwork in cutting & filling.

UNIT-3: MODERN FIELD SURVEY SYSTEMS **14 Hrs.**

Principle of Electronic Distance Measurement, Modulation, Types of EDM instrument modern surveying equipment, Total Station - Parts, Accessories, Advantages and Applications, land surveying with total stations, Field Procedure for total station survey, Errors in Total Station Survey; Global Positioning Systems- Segments, GPS measurements, errors and biases, Surveying with GPS, Co-ordinate transformation, accuracy considerations.

UNIT-4: ADVANCE SURVEYING TECHNIQUES **14 Hrs.**

Principles of Remote Sensing, Electromagnetic Spectrum, interaction of electromagnetic radiation with the atmosphere, sensors and platforms, concept of resolution, photogrammetry, drone and UAV survey, GIS introduction, GIS data types, GIS data sources, cartography, GIS applications, Introduction of QGIS in Advanced surveying, applications of GIS in Civil Engineering.

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES:

CO1 : Define the basic principles of linear, angular, and levelling
CO2 : Illustrate the application of theodolite, auto level, and total station.
CO3 : Examine and measure the area and volume of a given field.
CO4 : Demonstrate the basic working of Tacheometry.
CO5 : Analyze the uses of modern tools and mapping methods in survey projects.
CO6 : Develop the working knowledge of Remote Sensing, GIS, and GPS for Surveying and mapping

TEXT-BOOK AND REFERENCE BOOKS:

1. NN Basak, **"Surveying and Levelling"**, 2nd Edition, Tata McGraw Hill Publication, New Delhi.
2. BC Punmia, "**Surveying Vol. I"**, 17th Edition, Laxmi Publishing House, New Delhi
3. Madhu, N, Sathikumar, R and Satheesh Gobi, **"Advanced Surveying: Total Station, GIS and Remote Sensing"**, Pearson India, 2006.
4. Haywood L, Cornelius S and S Carver, "**An Introduction to Geographical Information Systems"**, Addison Wiley Longmont, New York.
5. Anji Reddy, M., "**Remote sensing and Geographical information system"**, B.S. Publications, 2001

					Surveying Practices Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	02	--	--	--	50	50	100

COURSE OBJECTIVES

1. To learn the usefulness of traditional survey methods through chain, tape, and compass survey
2. To gain knowledge by learning theodolite and total station for elevation calculation and contours.
3. To learn GPS and GPS based data point mapping for engineering.
4. To understand the GIS application in civil engineering survey and mapping.

LIST OF EXPERIMENTS

1 To conduct the linear measurement using the chain and offset method
2 To measure a linear and angular distance using Chain and Compass Survey
3 To conduct profile levelling along a road.
4 To prepare contour maps by Grid methods
5 To measurement the horizontal and vertical angles with total station
6 To determining the area of irregular boundary with planimeter.
7 To setting out a simple circular curve.
8 To conduct the advance survey using Global Positioning System (Point, Line, and Area)
9 To study the advance software techniques for map preparation with GIS.
10 Project 1: Calculating Cross-sectional Areas of Road Formations.
11 Project 2: Computing Volumetric Quantities of Earthwork along a Road Alignment
12 Project 3: Catchment delineation with QGIS
13 Project 4: Drone survey and data processing
14 Project 5: Drone flying simulator.

COURSE OUTCOMES:

CO1 : Explain the usefulness of traditional survey methods through chain, tape, and compass.
CO2 : Make use of Auto level for levelling survey.
CO3 : Determining vertical and horizontal angles with theodolite/total station.
CO4 : Design and layout the simple circular curve on field.
CO5 : Illustrate the application of GPS and GIS on field survey.
CO6 : Demonstration mapmaking skills with GIS data.

TEXT-BOOK AND REFERENCE BOOKS:

1. NN Basak, **"Surveying and Levelling"**, 2nd Edition, Tata McGraw Hill Publication, New Delhi.
2. Madhu, N, Sathikumar, R and Satheesh Gobi, **"Advanced Surveying: Total Station, GIS and Remote Sensing"**, Pearson India, 2006.
3. Haywood L, Cornelius S and S Carver, "**An Introduction to Geographical Information Systems"**, Addison Wiley Longmont, New York.

					Structural Analysis					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs / Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

1. Attain proficiency in employing various structural analysis methods to solve complex engineering problems.
2. Apply fundamental theorems and concepts of structural analysis to assess the behavior of different structural systems.
3. Enhance problem-solving abilities by analyzing and interpreting structural responses under different loading conditions.

UNIT I: Strain Energy, Deflections and Two hinged arches **13 Hrs.**

Stability and Determinacy of Structure Statically determinate and indeterminate structures by force/ energy methods; Method of superposition, Plane Trusses, Deflection of Trusses. Deflections of beam and frame. Introduction to arches.

UNIT II: ILD, moving loads and consistent deformation **13 Hrs.**

Influence Line Diagrams for reaction, shear force and bending moments of statically determinate beams. Maximum bending moments and shear forces at specified sections due to static and moving loads. Consistent deformation method/Method of least work: Analysis of single span beams with various loading and boundary conditions

UNIT III: Slope deflection method **15 Hrs.**

Fixed beams analysis for end moments and introduction to continuous beams, Slope deflection method: Introduction, Sign convention, Development of slope-deflection equations and Analysis of beams and orthogonal rigid jointed plane frames (non-sway and sway analysis)

UNIT IV: Moment distribution method **15 Hrs.**

Moment Distribution Method: Introduction, Definition of terms-Distribution factor, Carry over factor, Development of method and Analysis of beams and orthogonal rigid jointed plane frames (non-sway and sway analysis)

Total Hours 56 Hrs.

COURSE OUTCOMES:

CO-1: Determine slopes and defection using strain energy, unit load method
CO-2: Analyze beams and arches and trusses using Castigianos theorems, method of least work or consistent deformation
CO-3: Draw ILD for statically determinate beam
CO-4: Compute max BM and SF for moving loads
CO-5: Analyze the statically indeterminate beams and simple frames using Slope deflection method
CO-6 Analyze the statically indeterminate beams and simple frames using moment distribution method

TEXT-BOOK AND REFERENCE BOOKS:

1. Reddy C. S., **"Basic Structural Analysis"**, Tata McGraw Hill, New Delhi.
2. R.C.Hibbeler, **"Structural Analysis, sixth Edition"**, Pearson publications, New Dehli
3. S.B.Junnarkar & H J Shah, **"Mechanics of Structures"**, Volume I, Charotar publishing house, Anand
4. Thandava Murthy, **"Analysis of Structures"**, Oxford University Press.
5. S Ramamrutham, **"Theory of structures"**, Dhanpat rai Publications, New Delhi.